

AI

Artificial Intelligence:- AI is the study of how to make computer do things which people do better. [machine + human Intelligence]

↳ AI can cause a machine to work as human.

↳ AI → Artificial [Man-Made]
 → Intelligence [Power of thinking]

GOALS OF AI:

- i) Replication of Human Intelligence.
- ii) Solving problems that require knowledge.
- iii) Building a machine that can do human Intelligence task. [CHESS, Proving theorem, automated car driving...]
- iv) Providing advise to the user.
- v) Intelligent connⁿ b/w a perception and connⁿ.

Reasons of Boost in AI:

↳ i) S/W or device can be made to solve Real-time Problems.

ii) Creation of Virtual assistant [SIRI, CORTANA]

iii) Robots development.

[Helps in dangerous env. condⁿ]

iv) New Job opportunities.

AI programming focuses on three cognitive aspects, such as learning, reasoning, and self-correction.

- **Learning Processes**
- **Reasoning Processes**
- **Self-correction Processes**

1. Learning Processes

- This part of AI programming is concerned with **gathering data and creating rules** for transforming it into useful information.
- The rules, which are also called algorithms, offer computing devices with step-by-step instructions for accomplishing a particular job.

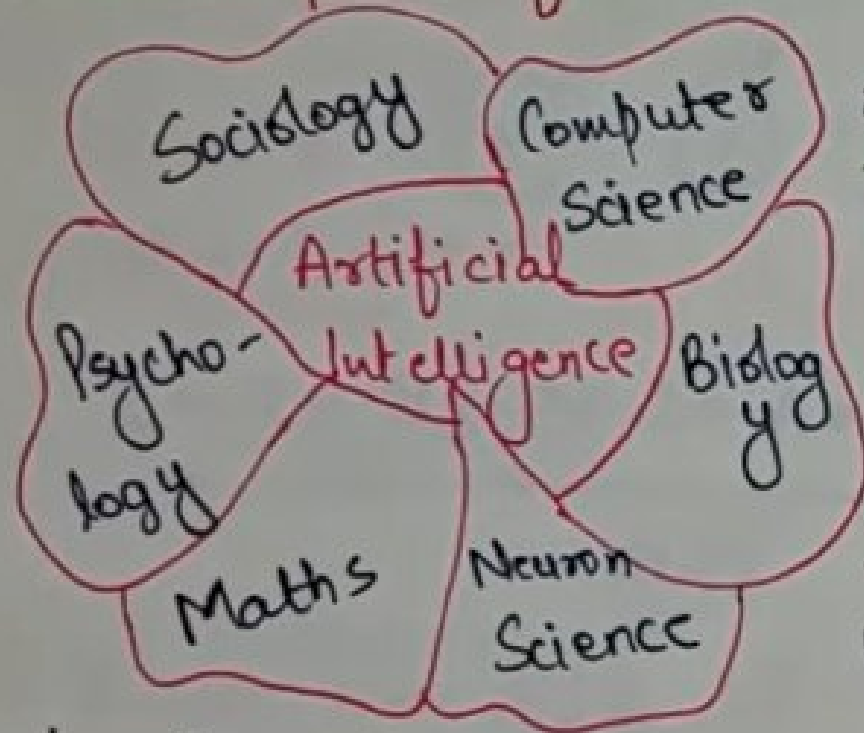
2.Reasoning Processes

This part of AI programming is concerned with selecting **the best algorithm** to achieve the desired result.

3.Self-Correction Processes

This part of AI programming aims to do **fine-tuning of algorithms** regularly in order to ensure that they offer the most reliable results possible.

AI is Comprised of:-



- ↳ Reasoning
- ↳ Learning
- ↳ Problem solving
- ↳ Language Understanding.

Advantages of AI

Accuracy $\uparrow$ & Error $\downarrow$

Fast Decision Making.

Reliability is more

usefulness in Risky Area.

Digital Assistant

Disadvantages of AI

COST $\uparrow$

Can't think beyond the limits.

No feelings & emotions.

more dependency on machines $\uparrow$.

No original thinking.

Applications of AI:-

- ① AI in Gaming: Chess, Poker, tic-toe.
↳ machine can think large no. of moves.
- ② AI in NLP: Natural lang. Processing
↳ Machine can understand human lang.
- ③ AI in Healthcare: Fast diagnosis
↳ Robotic Surgery.
- ④ AI in Finance: Adaptive Intelligence.
↳ automatic chatbots, algorithm trading.
- ⑤ AI in Data Security: Helps in making data/appⁿ more secure.
↳ AEG bot, AI2
- ⑥ Expert System: Integration of slw, machine and special infoⁿ to provide reasoning & advise.
- ⑦ Computer Vision: Understand the visual automatically by machine.
- ⑧ Speech Recognition: Extract - the meaning of sentence by human talk. [slang removal, noise rem. -]
- ⑨ Robotics: Talk and behave like humans. ↳ Erica and Sophia.
- ⑩ AI in e-Commerce: Automatic recommendation.

Classification of AI

Classification of AI:-

Narrow AI

Narrow AI: Able to perform dedicated task with intelligence. [Not concerned with How]
↳ Can't perform beyond its field or limitations.

↳ Example:-

- ↳ Flying machine
- ↳ Using logics
- ↳ Apple SIRI
- ↳ Playing chess

Evolutionary AI: It is the study and design of machines that simulate simple creatures and attempt to evolve.

↳ Example: Ants, Bees etc.

Strong AI: It is the study and design of machines that simulate human mind to perform intelligent tasks.

↳ ii) Borrowing ideas from psychology and neuroscience

↳ iii) Forgetting things, Genetics, Language.

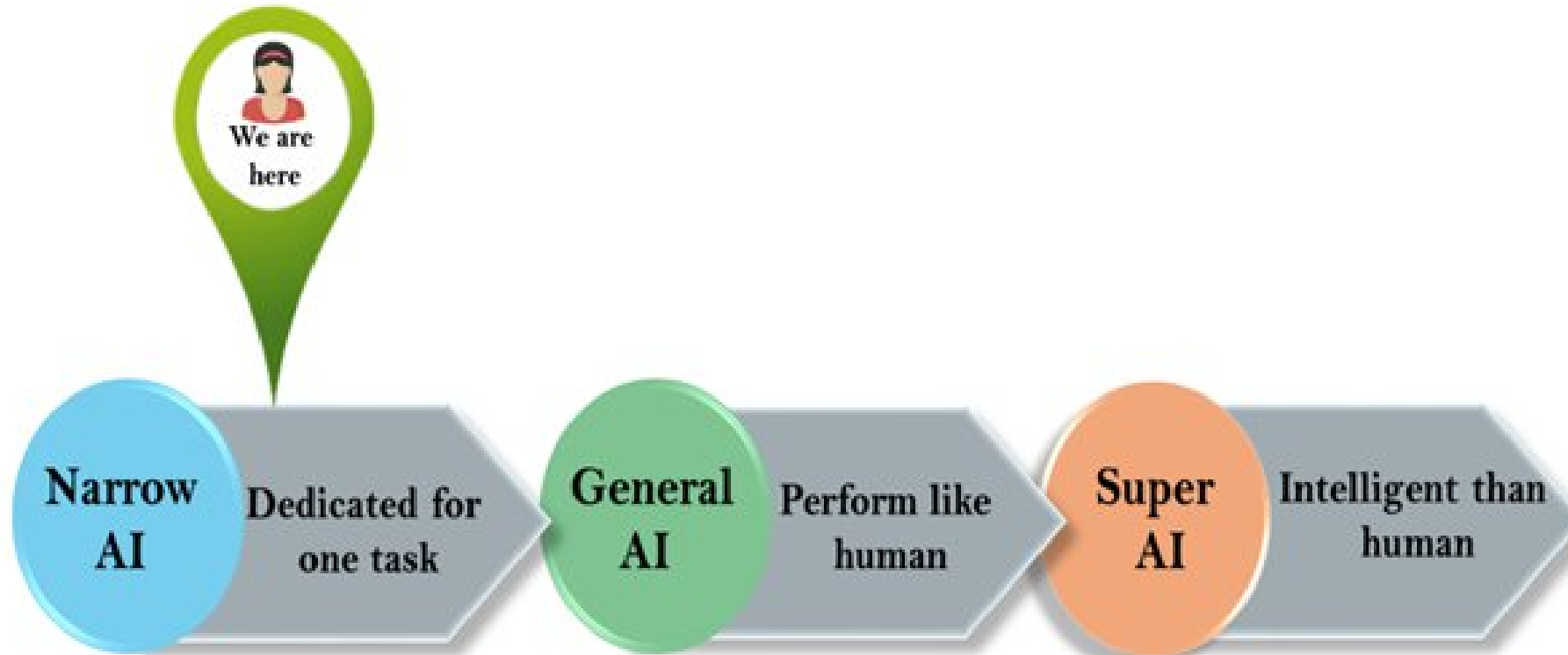
Super AI:- Hypothetical concept.
machine > Human.
[machine ↔ machine]

2. General AI:

- General AI is a type of intelligence which **could perform any intellectual task** with efficiency like a human.
- The idea behind the general AI to make such a system which **could be smarter and think like a human** by its own.
- Currently, there is **no such system exist** which could come under **general AI** and can perform any task as perfect as a human.
- The worldwide researchers are now focused on developing machines with General AI.
- As systems with **general AI are still under research**, and it will take lots of efforts and time to develop such systems.

3.Super AI:

- Super AI is a level of Intelligence of Systems at which **machines could surpass human intelligence**, and can perform any **task better than human** with cognitive properties. It is an outcome of general AI.
- Some key **characteristics of strong AI include** capability include the ability to think, to reason, solve the puzzle, make judgments, plan, learn, and communicate by its own.
- Super AI is still a **hypothetical concept of Artificial Intelligence**. Development of such systems in real is still world changing task.



- **Artificial Intelligence is a vast field and Machine Learning is a subset of Artificial Intelligence and Deep Learning is subset of Machine Learning.**



- **Introduction of AI**
- AI History and applications, Overview of AI application areas, Depth First Search, Breadth First search, game playing, automated reasoning and theorem proving, expert systems, natural language understanding, planning and robotics, machine learning and Alan Turing Test, Intelligent Agents.

[8Hrs]

History of Artificial Intelligence

- **1940s–1950s: Foundations**
- **Alan Turing** proposed the idea of machines that could simulate human intelligence. His 1950 paper introduced the **Turing Test**.
- In 1956, the term "**Artificial Intelligence**" was coined at the **Dartmouth Conference**, marking the formal birth of AI as a field.
- **1960s–1970s: Early Progress**
- Researchers built **rule-based systems** and **simple problem solvers**.
- Programs like **ELIZA** (a chatbot mimicking a psychotherapist) showed early promise in natural language processing.
- **1980s: Expert Systems**
- AI shifted toward **knowledge-based systems** that mimicked expert decision-making.
- These systems were widely used in medicine, finance, and engineering.

- **1990s–2000s: Machine Learning Emerges**
- Focus moved from hard-coded rules to **learning from data**.
- **IBM's Deep Blue** defeated chess champion Garry Kasparov in 1997, showcasing AI's strategic capabilities.
- **2010s–Present: Deep Learning Revolution**
- **Neural networks**, especially **deep learning**, transformed AI.
- Breakthroughs in **image recognition**, **speech processing**, and **natural language understanding** led to tools like **Siri**, **Google Translate**, and **ChatGPT**.
- AI became embedded in **smartphones**, **autonomous vehicles**, **healthcare diagnostics**, and **robotics**.

Applications of AI

-  **1. Healthcare**
 - **Medical imaging analysis, drug discovery, and predictive diagnostics.**
 - AI models detect diseases like cancer, diabetic retinopathy, and heart conditions.
- **2. Finance**
 - **Fraud detection, algorithmic trading, and credit scoring.**
 - AI helps banks assess risk and automate customer service.
- **3. Transportation**
 - **Self-driving cars, traffic prediction, and fleet optimization.**
 - Companies like Tesla and Waymo use AI for autonomous navigation.
- **4. Manufacturing**
 - **Predictive maintenance, quality control, and robotic automation.**
 - AI enhances productivity and reduces downtime.

- **5. Education**
- **Personalized learning platforms, automated grading, and intelligent tutoring systems.**
- AI adapts content to individual student needs.
- **6. Natural Language Processing**
- **Chatbots, language translation, sentiment analysis, and text summarization.**
- Used in customer support, social media monitoring, and content creation.
- **7. Agriculture**
- **Crop monitoring, yield prediction, and automated irrigation.**
- AI helps optimize resource use and improve food security.
- **8. Cybersecurity**
- **Threat detection, anomaly analysis, and automated response systems.**
- AI defends against evolving digital threats.

Ques:- what do you mean by Turing Test? How it can be performed?

Coined by Alan Turing in 1950.

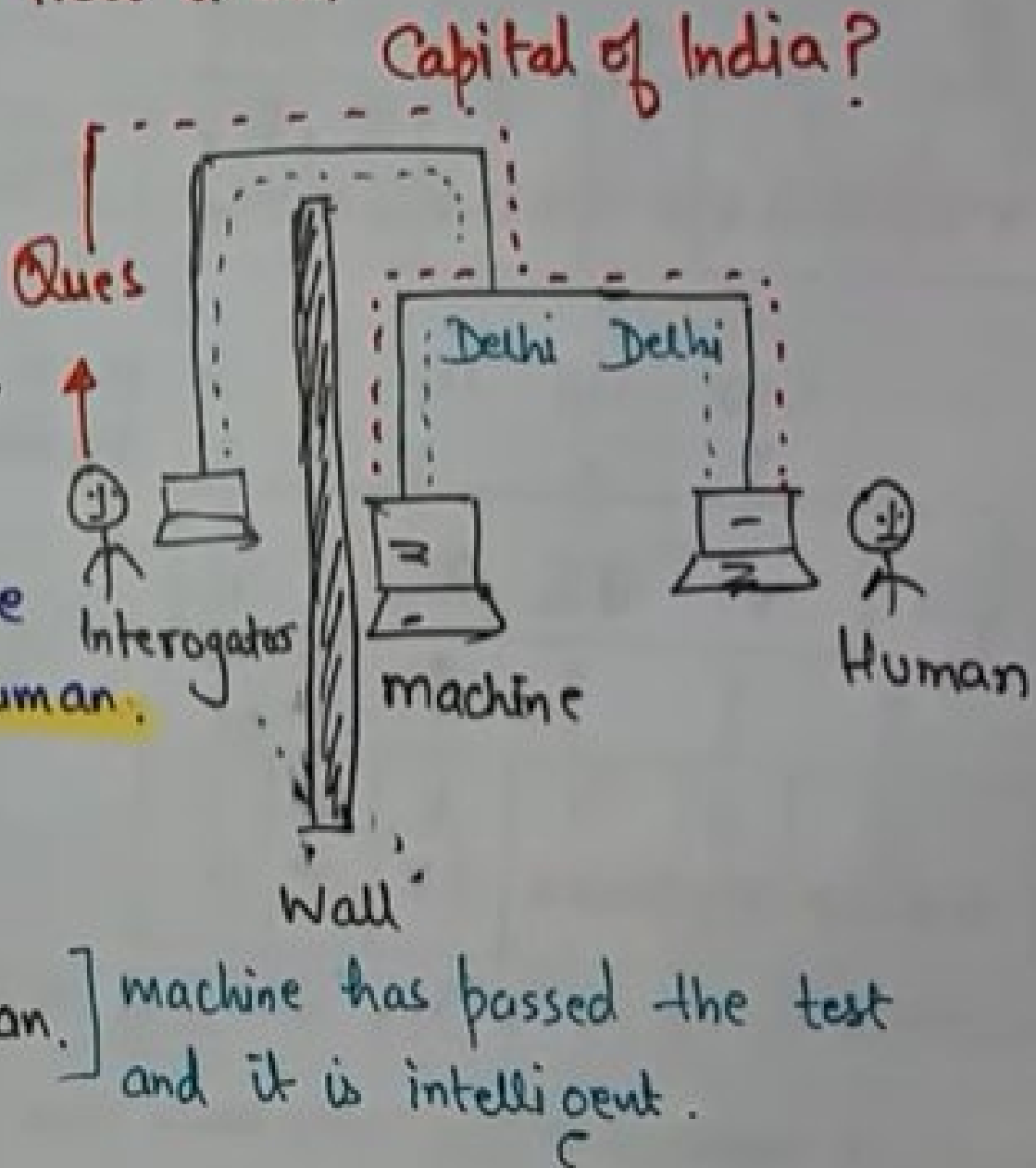
"Turing test is used to determine 'whether or not machines can think intelligently like humans'."

Basic Configuration:-

→ There will be a human interrogator on one side of wall and other side a machine and human.

→ Machine intelligent ⇒ when human interrogator can't distinguish response given by machine and human.

→ Knowledge Base.



machine has passed the test and it is intelligent.

<https://www.youtube.com/watch?v=dWpvK-8uAqM>

↳ Recursive algorithm.

↳ Implemented using **STACK** (LIFO)

i) Enter Root node on Stack

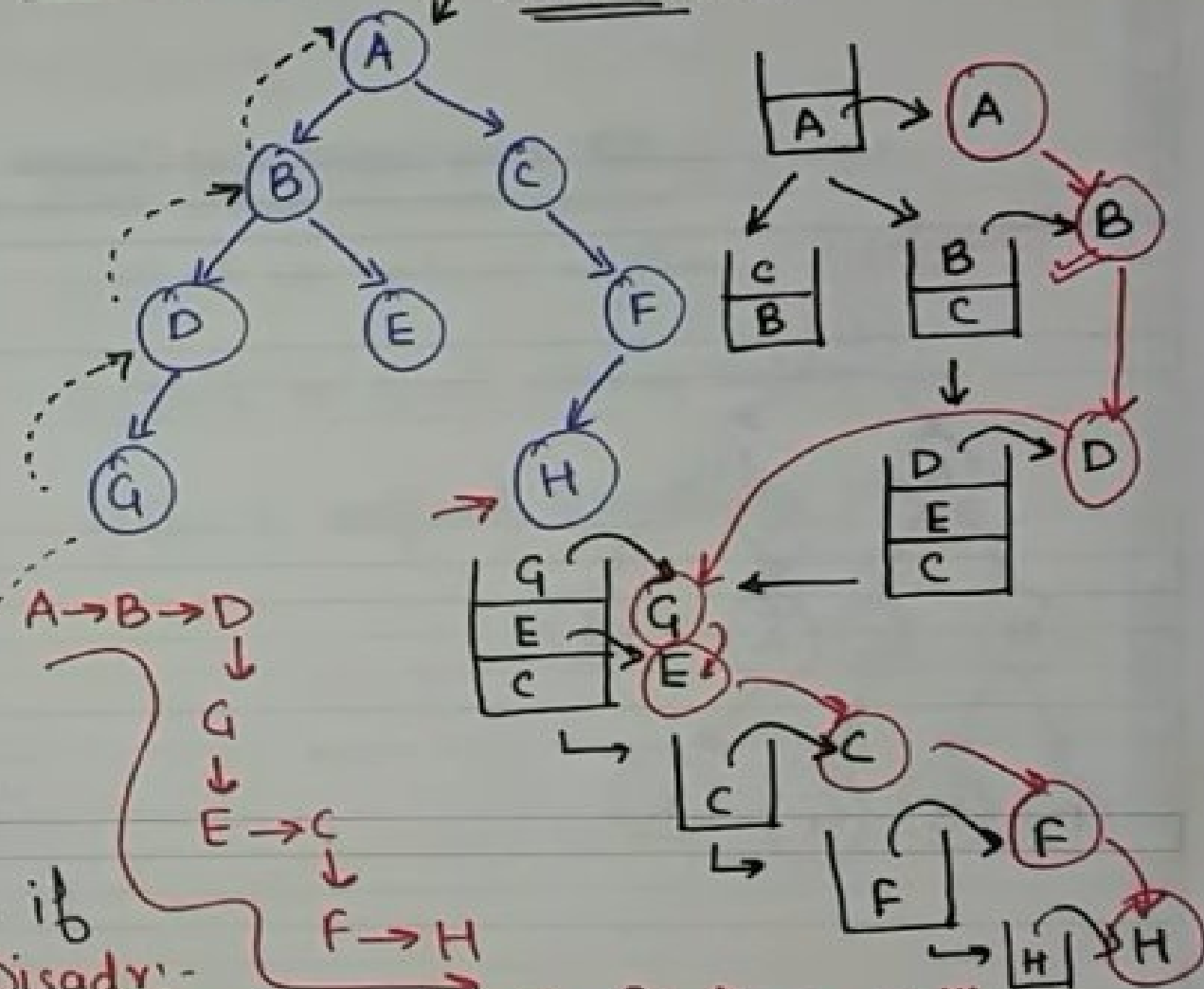
↳ a) Remove Node ^{→ (POP)}

ii) Push ^{children of} all node in stack

↳ Less time to reach goal node if traversal in right path. | Disa

- ↳ No Guarantee of finding solⁿ.
- ↳ Infinite loop.

Start node.



Breadth First Search

<https://www.youtube.com/watch?v=SZ2a668G6vA>

BFS:- Breadth First Search.

↳ Explores all the nodes at given depth before proceeding to the next level.

↳ Uses Queue to implement.

ALGORITHM:

i) Enter Starting nodes on Queue.

ii) If Queue is empty, then return fail and stop.

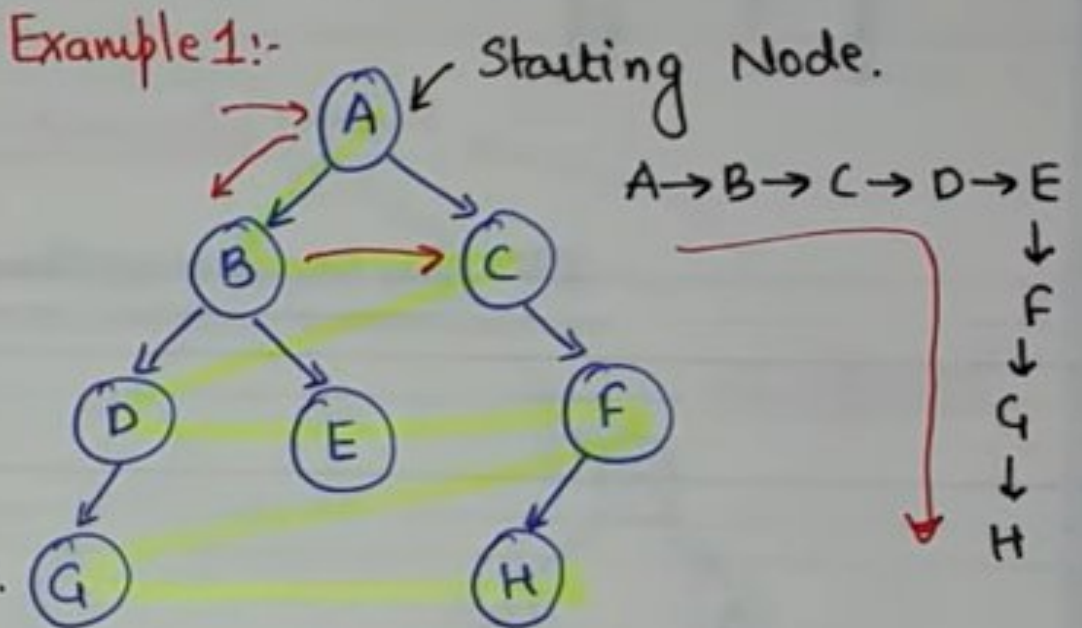
iii) If first element on Queue is Goal Node, then return Success and stop.

ELSE

(IMP)
iv) Remove and expand first element from Queue and place children at end of Queue.

v) Goto Step (ii)

Example 1:-



Initial Queue {A} = Goal node x

{B, C} ↓ (D, E)

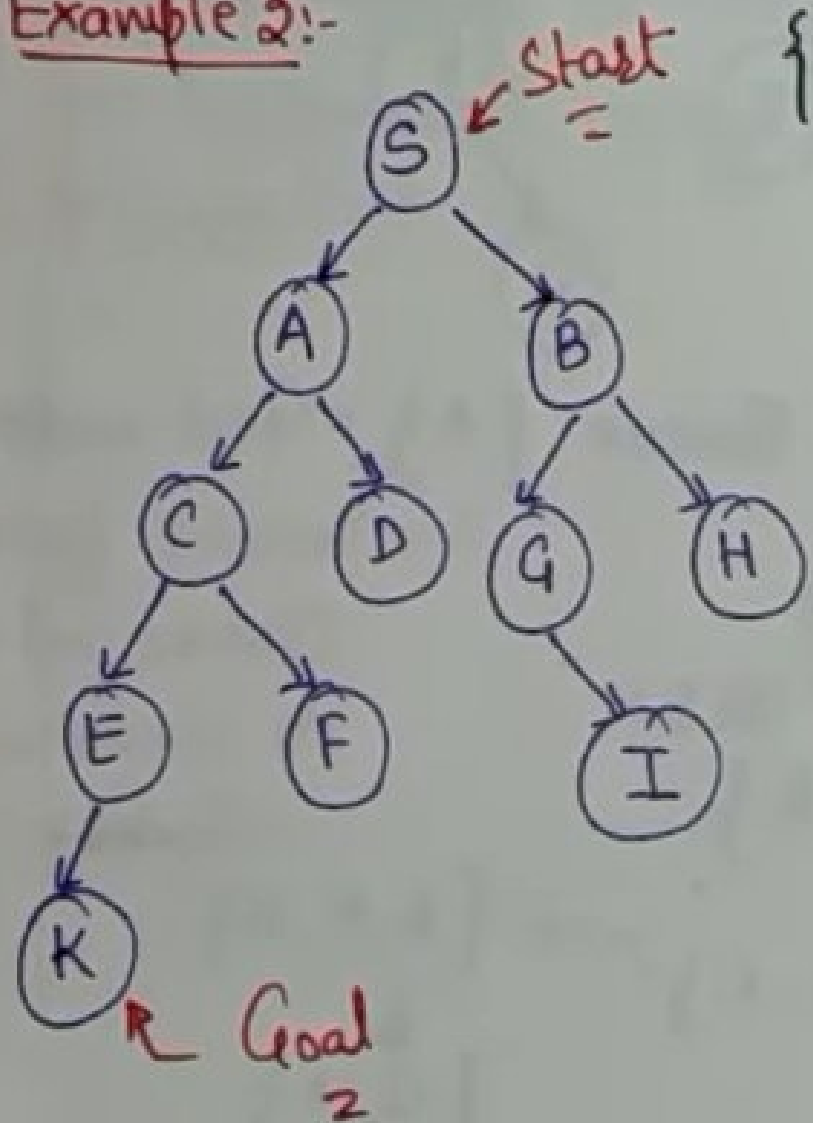
{C, D, E}

{D, E, F} → {E, F, G}

{G, H}

↳ Remove from Queue and add its successor to Queue.

Example 2:-



$\{S\}$

$\hookrightarrow \{A, B\}$

$\hookrightarrow \{B, C, D\}$

$\hookrightarrow \{C, D, G, H\}$

$\hookrightarrow \{D, G, H, E, F\}$

$\hookrightarrow \{H, E, F, I\}$

$\hookrightarrow \{F, I, K\}$

$G \rightarrow H \rightarrow E \rightarrow F \rightarrow I \rightarrow K \hookrightarrow$

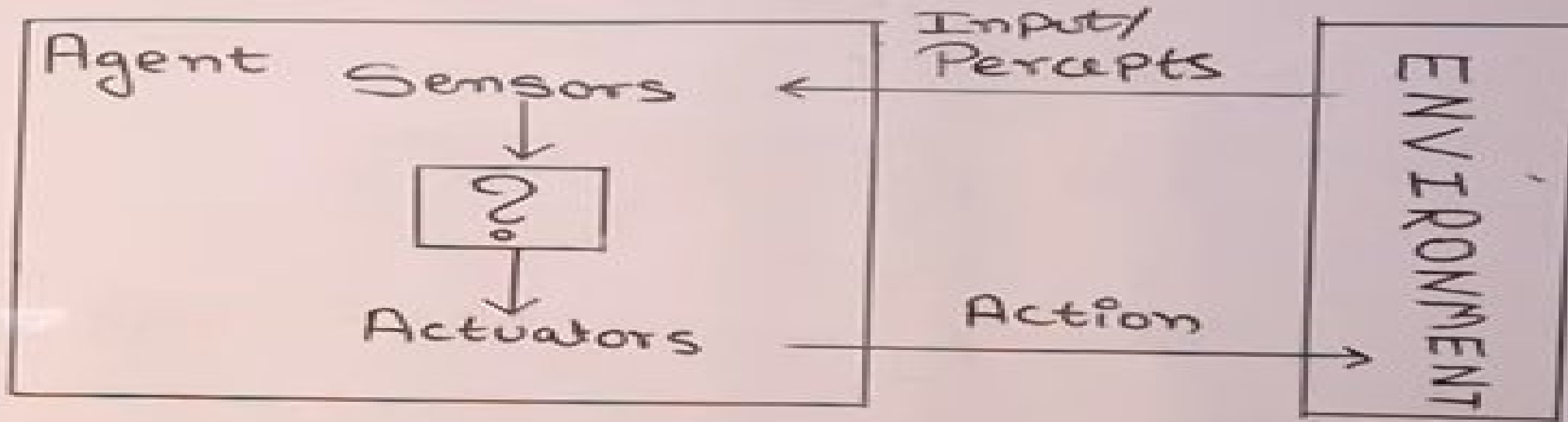
Adv:- find a solⁿ if its exists.
 $\hookrightarrow$ minimal solⁿ in least no. of steps.

Disadv:- More memory

$\hookrightarrow$ need lots of time

$\hookrightarrow$ solⁿ is far from root node.

AI AGENT



Agent is something that perceives its Environment through its Sensors and act upon that Environment through its Actuators.

Robotic Agent:

Sensors: Camera etc

Actuators: Legs, Hands etc

PEAS: Grouping of AI Agents:

↳ used to group similar type of agents together.

↳ PEAS: Sensor: Devices from which agent perceives observation from env

↳ Actuators: devices, H/w, S/w through which Agent performs action on env.

↳ Performance measure: Environment
↳ All surrounding things and conditions.

It is the op we get from an Agent.

↳ Results obtained after Agent Processing.

Example:-

SELF DRIVING CAR

(P):- Comfort, Safety, Time, Legal driving.

(E):- Condⁿ of Roads, Crossings, Traffic signals.

(A):- Steering, Brakes, Horn, Accelerator.

(S):- Camera, GPS,

Example of AI Agent

- 1) Human Agent
- 2) Robotic Agent
- 3) Software Agent



Sensors, Effectors Percepts and Actions

■ Humans

- **Sensors:** Eyes (vision), ears (hearing), skin (touch), tongue (gustation), nose (olfaction), neuromuscular system (proprioception)
- **Effectors:** limbs, eyes, tongue, ...
- **Percepts:**
 - At the lowest level – electrical signals from these sensors
 - After preprocessing – objects in the visual field (location, textures, colors, ...), auditory streams (pitch, loudness, direction), ...
- **Actions:** lift a finger, turn left, walk, run, carry an object,

Types of AI Agent

1. Simple Reflex Agent

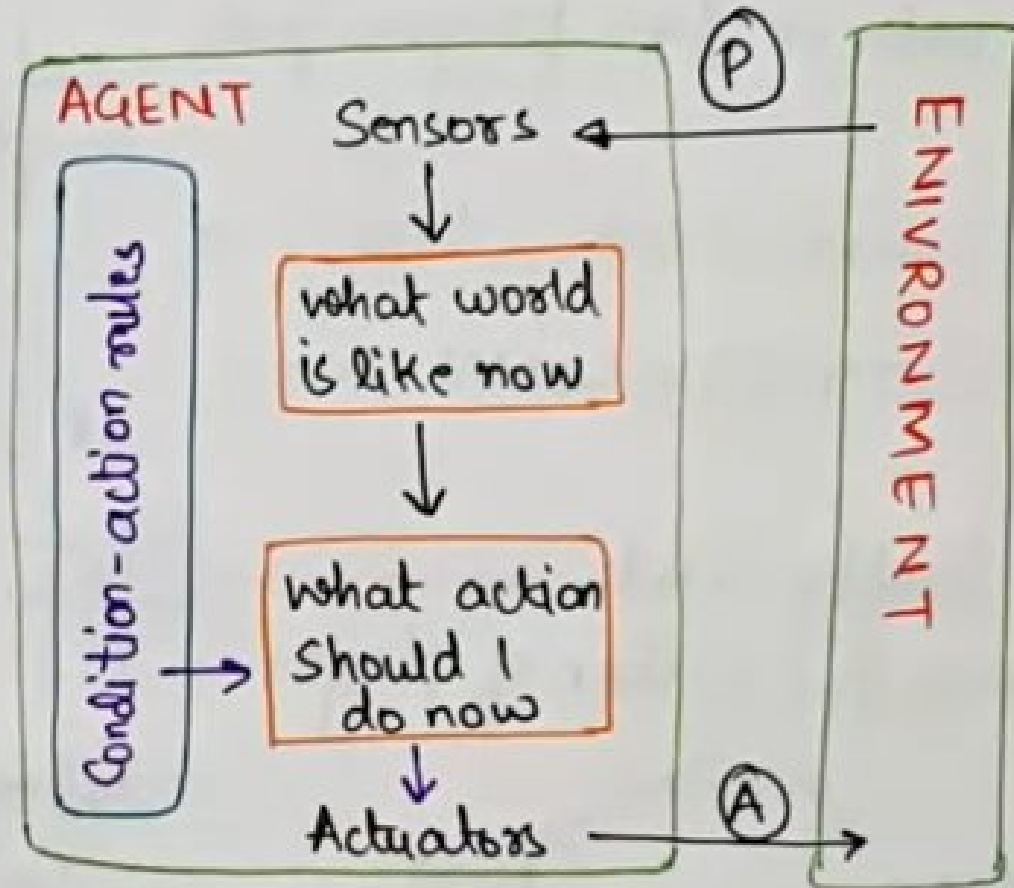
① Simple Reflex Agents:

Works only on current situation/perception and ignores the history of previous state. True

↳ Condition-Action Rule

Limitations:-

- ↳ i) Very limited Intelligence.
- ii) No Knowledge about non-perceptual parts of state
- iii) Can go into infinite loop.



② ~~Model~~ Model based Reflex Agent:

↳ Works by finding the rule whose Condⁿ matches Current Situation.

↳ Can work in partially observable env., and track situation.

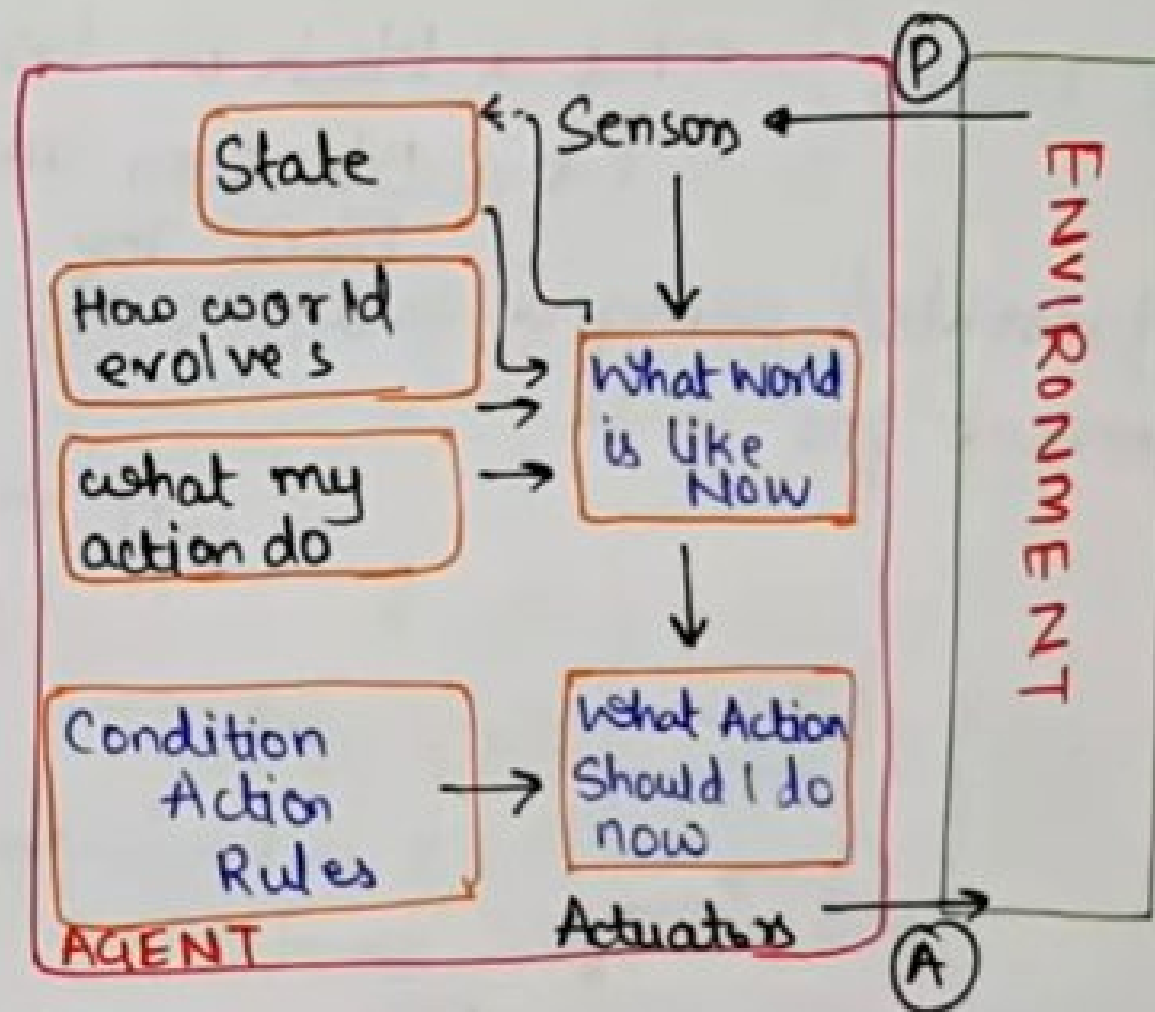
↳ Agent keeps track of internal state which is adjusted by each percept and that depends on percept history.

↳ Model: How things happen in world.

↳ Agent State updation

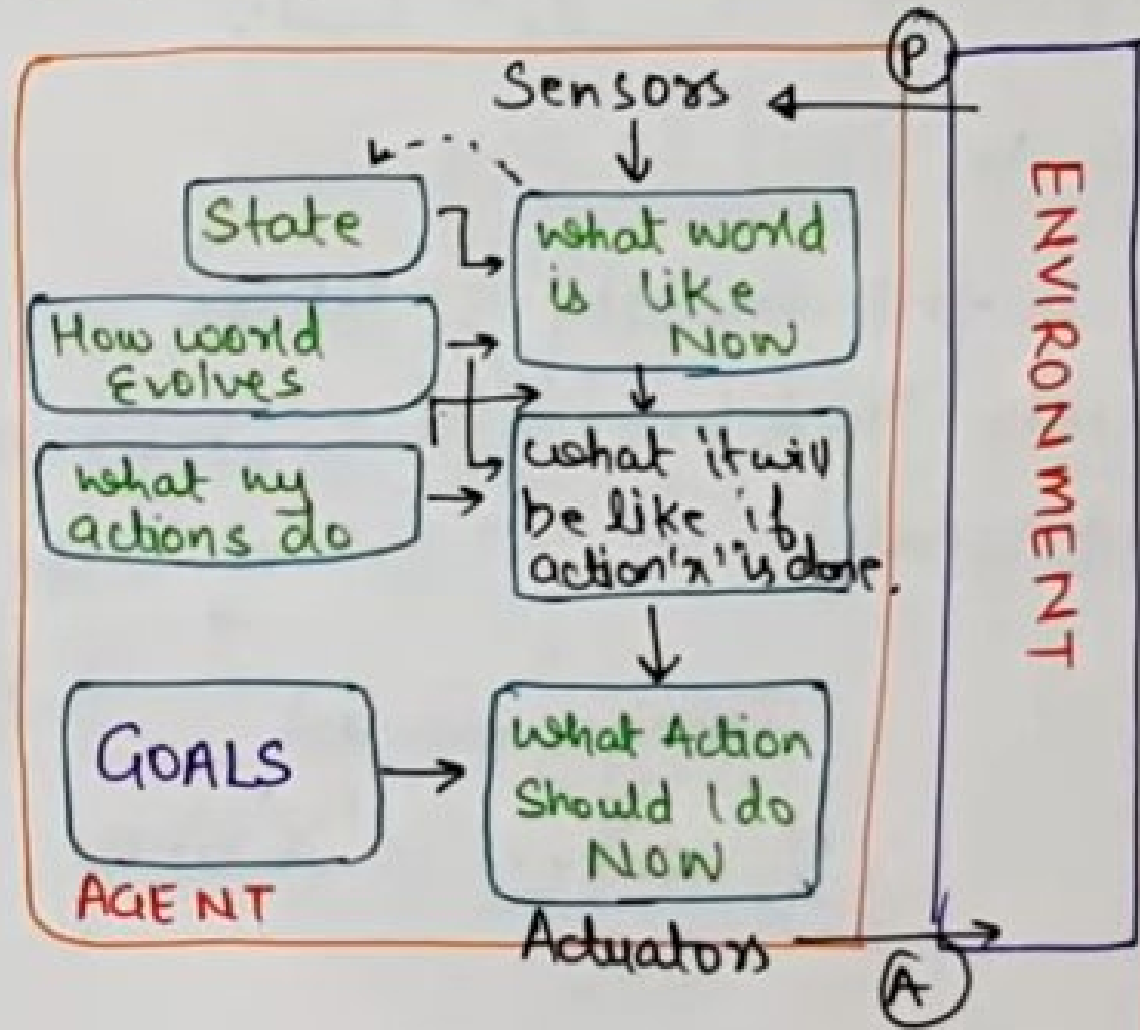
required Infoⁿ:

- ① How world is evolving
- ② How agent's action affect the world.



③ Goal-Based Agents:

- ↳ Focuses only on reaching the goal set.
- ↳ Agent takes decision based on how far it is currently from the Goal State.
- ↳ Every action is taken to minimize distance to Goal State.
- ↳ More flexible Agent.

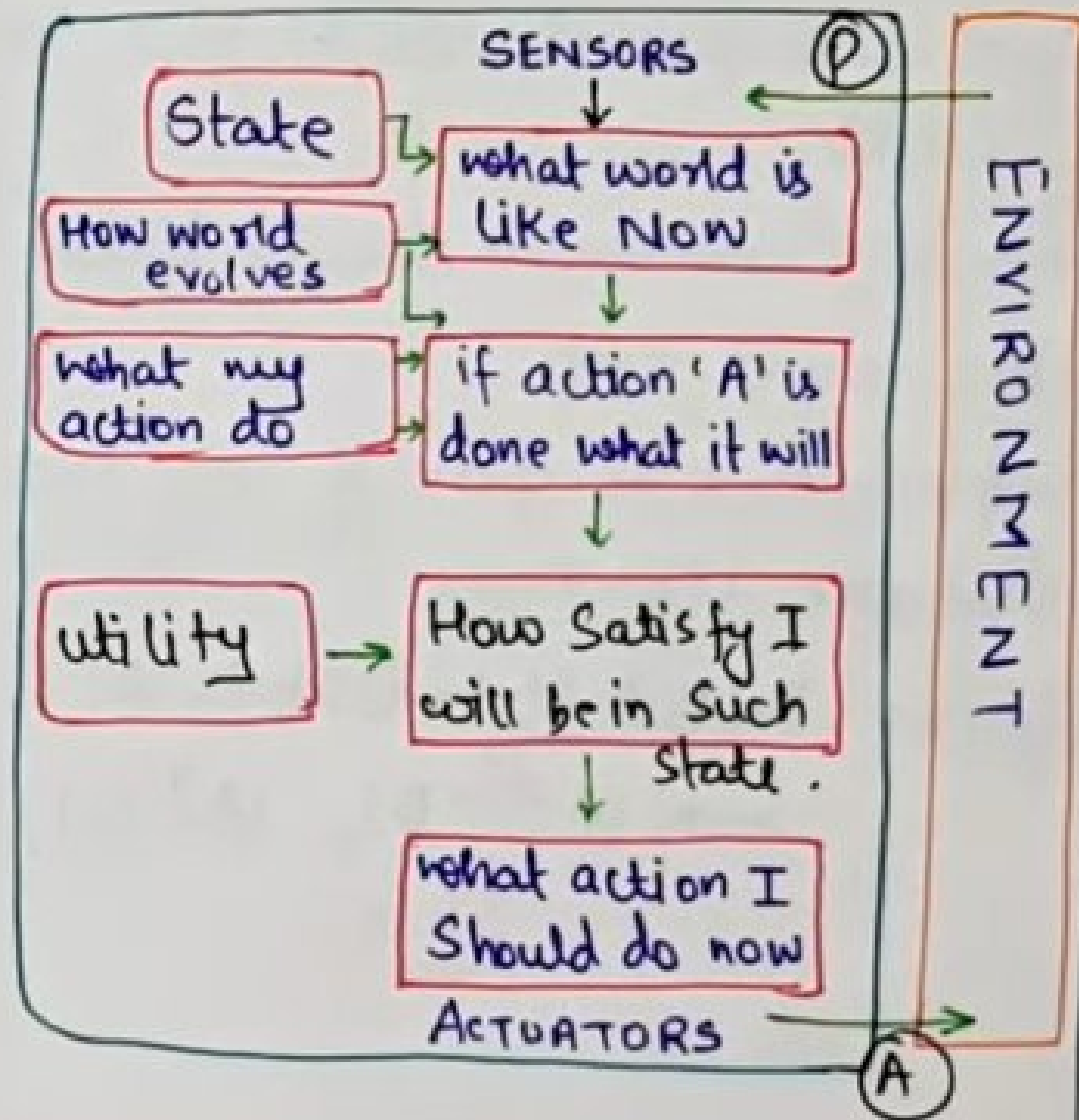
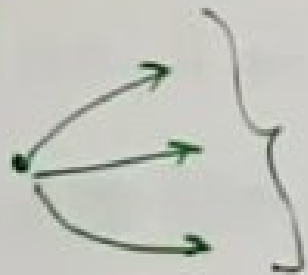


④ Utility-Based Agents:-

→ Agents are more concerned about the utility (Preference) for each state.

→ Act based not only on goals but also the best way to achieve goal.

→ Useful when there are multiple possible alternatives and agent has to choose in order to perform best action.



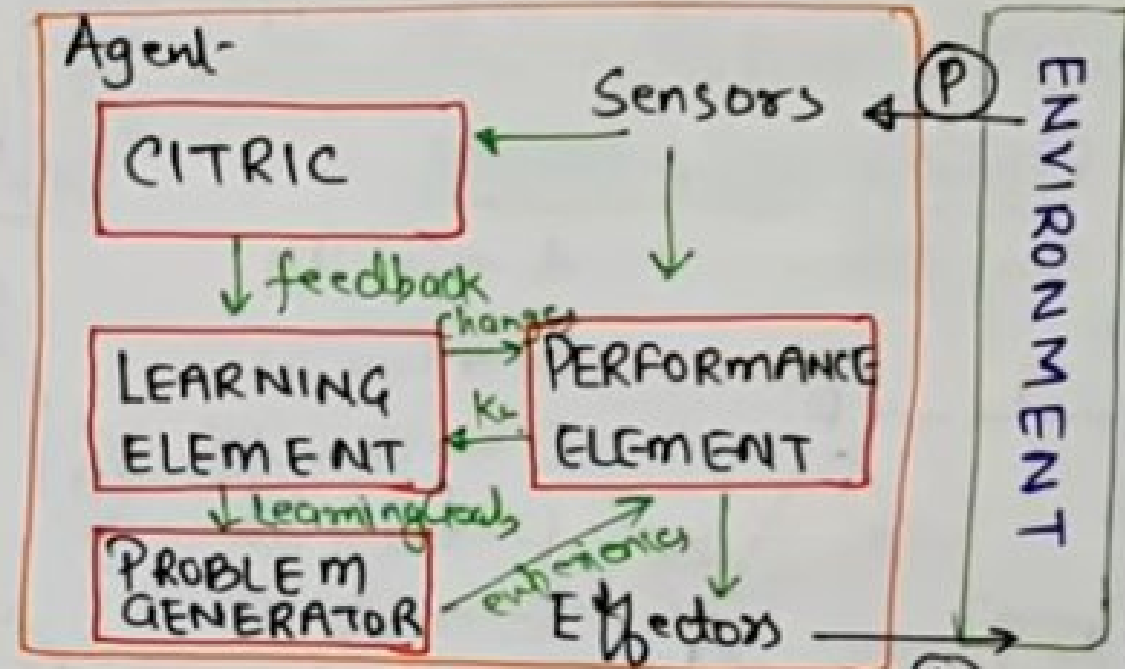
Example of Beam Search Algorithms

Learning Agents:-

- Can Learn from its past experiences.
- Starts to ACT with basic knowledge and then able to act by adapting learning.

Components:-

- (i) Learning Element: Makes Improvement in System by learning from env.
- (ii) Citric: Gives feedback about Agent's Performance based on standard.



- (iii) Performance Element: Selects the action - to Perform
- (iv) Problem Generator: Suggests the action.] new infoⁿ gain.

Ques:- Explain Min/Max theorem. [Game Playing]

It is a specialized Search Algo that returns optimal sequence of moves for a player in Zero sum Game.

↳ Recursive/Backtracking algo which is used in decision making and Game theory. → Two Player.

↳ uses recursion to search through Game Tree.

↳ Algo computes minimax decision for current state.

↳ Two Players

- MAX [Selects maximum Value]
- MIN [Selects minimum Value]

↳ Depth-First Search Algo is used for exploration of complete Game Tree.

↳ CHESS, checkers, Tic-tac-toe...

[max = $-\infty$] Worst Values.
[min = ∞] [Initial].

Properties:- → COMPLETE

① Definitely found solⁿ (if exists)

② optimal

③ Time Complexity = $O(b^m)$ ^{maximum depth.}
↳ branching factor of game tree

④ Space Complexity = $O(b^m)$

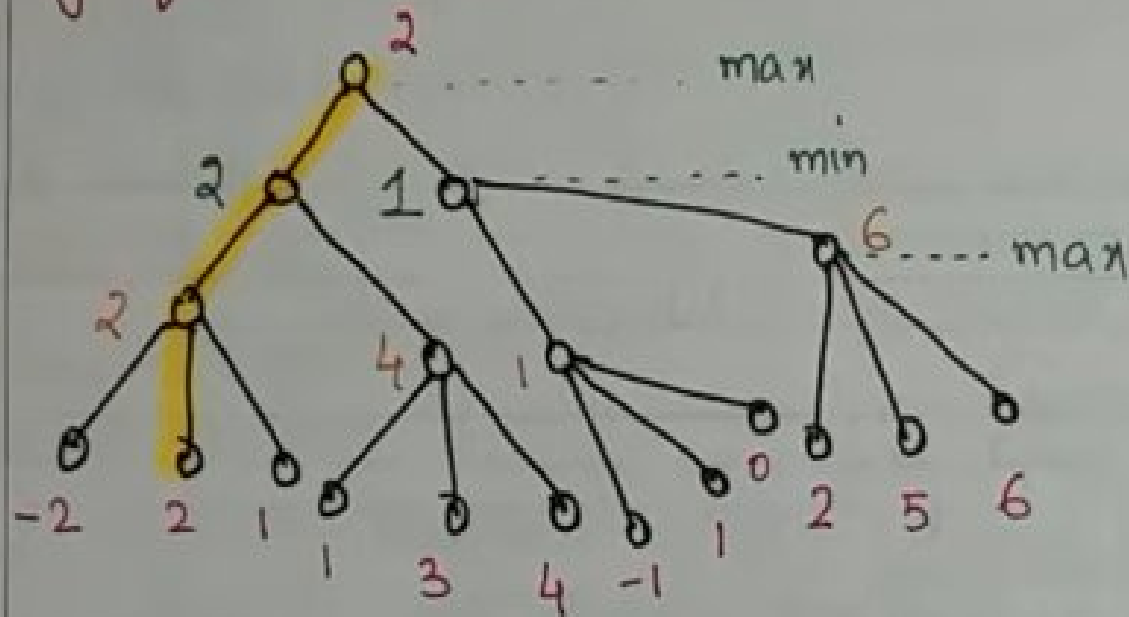
Limitations:-

↳ Slow for complex Games such as chess.

↳ 35 choices/moves.

$(35)^{100}$ d = 100 (for both players)
↳ BIG.

Eg. of MiniMax theorem:-



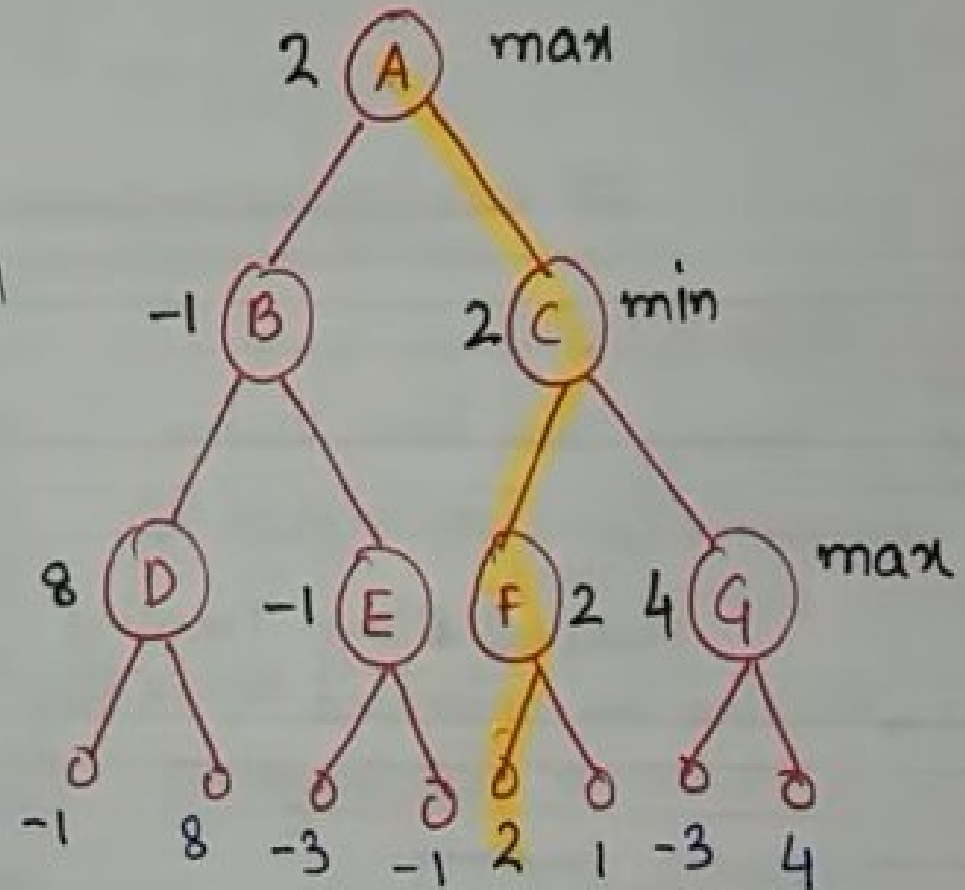
Initial Values $\left[\begin{matrix} \text{max} = -\infty \\ \text{min} = \infty \end{matrix} \right]$

$(-2, -\infty)$

$\rightarrow (-2, 2)$

$\rightarrow (2, 1)$

$\rightarrow (2)$



- <https://www.youtube.com/watch?v=S7L4-KDTvEE>

<https://www.youtube.com/watch?v=BkF-9yuiG18>

$$\max(4, -3) = 4$$

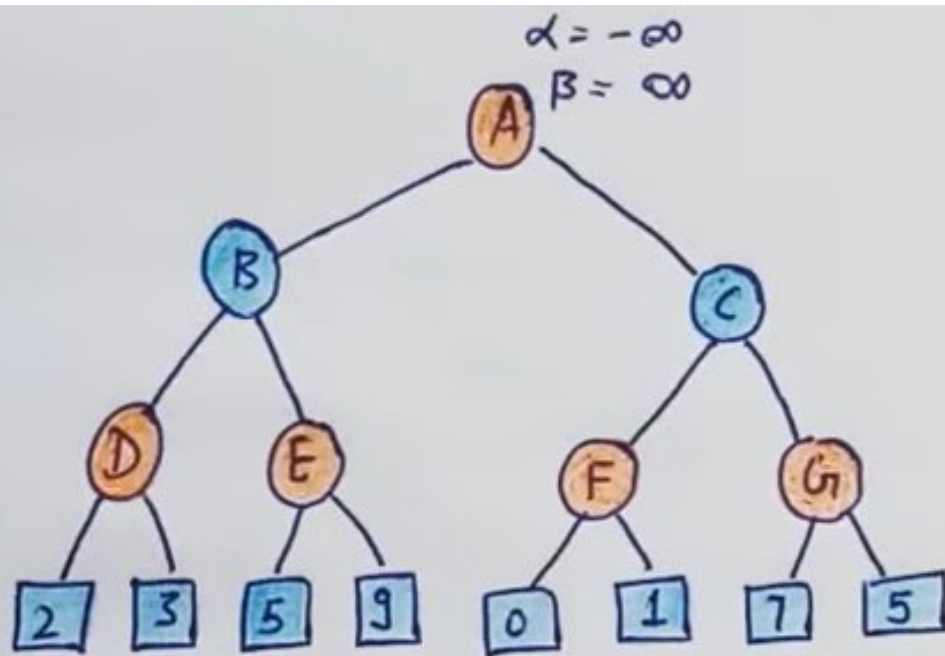
Alpha - Beta Pruning $\Rightarrow$

- It is a modified version of **minimax** algorithm. It is an optimization technique for minimax algorithm.
- There is a technique by which without checking **each node** of game tree we can compute correct minimax decision and this tech. is called **pruning**. This involves 2 threshold parameters **Alpha & beta** for future expansion. so it is called **Alpha - Beta pruning**.

Alpha :- The best (highest value) choice we have found so far at any point along the path of maximizer. (Max player)
The initial value of alpha is $-\infty$.

Beta :- The best (lowest value) ——— of minimizer. $+\infty$. (Min player)

→ It removes all the nodes which are not really affecting the final decision.



→ max

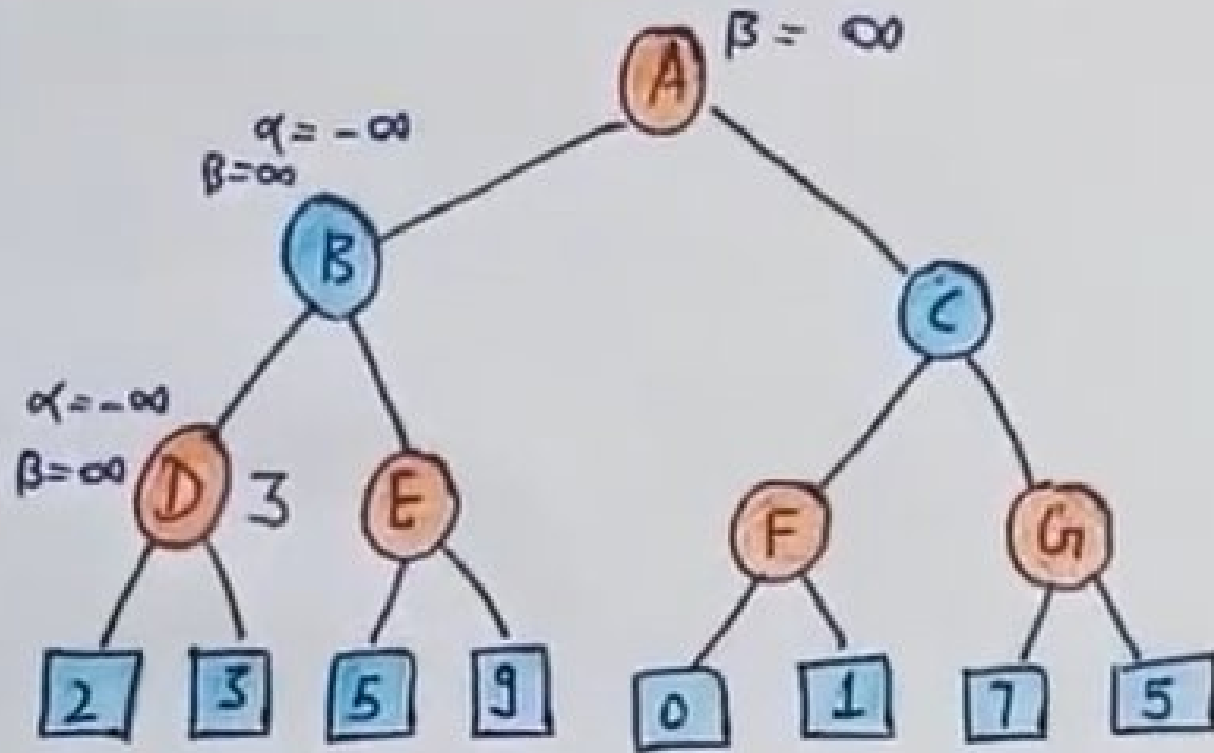
→ min

→ Max

→ Terminal node

Alpha Beta Algorithm

- while Backtracking, node values will be passed to upper nodes instead of values of alpha & beta.
- We will only pass alpha, beta values to child nodes.



→ max

→ min

→ Max

→ Terminal node

Alpha Beta
Algorithm...

(AI-12)

Lexicon: all the words and phrases used in a particular language or subject; all the words and phrases used and known by a particular person or group of people

Ques:- Explain the Various Components of Natural Language

Processing.

NLP: Branch of AI that helps computers to understand, interpret and manipulate human language.

→ Following tasks that can be done are:-

- ① Translation
- ② Summarization
- ③ Speech Recognition
- ④ Sentiment Analysis

(S.A) → Assign meanings

→ Shows how words are associated with each other.

(P.A) → Overall Communicative &

Social contents and its effect on interpretation

(m.p) → Vocabulary
that includes words and expressions
↳ depicts word str. info

↳ divides text into paragraphs, words, sentence.

derives meaningful use

Components:

rules, principles that governs the sentence structure of individual lang.

Lexicon

Grammar

Input Sentence

Morphological Processing

Syntax analysis

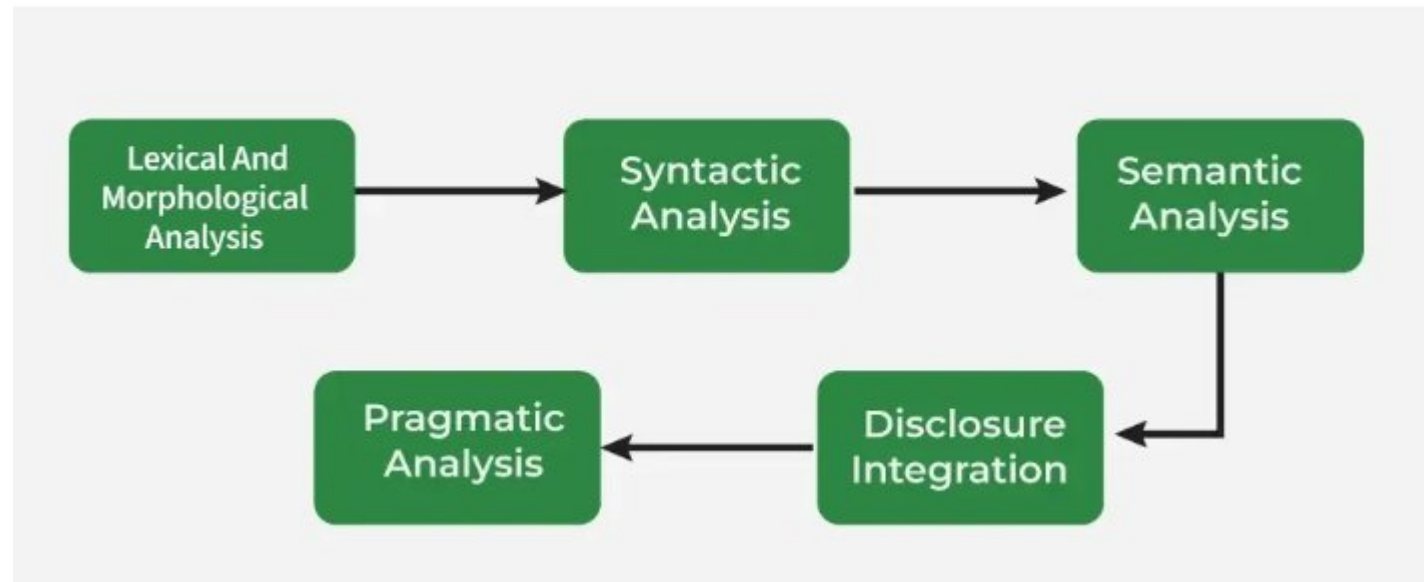
Semantic Rules

Semantic Analysis

Contentual Info

Pragmatic Analysis

- Natural Language Processing (NLP) helps computers to understand, analyze and interact with human language. It involves **a series of phases that work together to process language and each phase helps in understanding structure and meaning of human language.** In this article, we will understand these phases.



- **Lexical Analysis**

- It focuses on identifying and processing words (or lexemes) in a text. It breaks down the input text into individual tokens that are meaningful units of language such as words or phrases.
- **Key tasks in Lexical analysis:**
- **Tokenization**: Process of dividing a text into smaller **chunks called tokens**. For example the sentence "I love programming" would be tokenized into ["I", "love", "programming"].
- **Part-of-Speech Tagging**: Assigning parts of speech such as **noun, verb, adjective to each token** in the sentence. This helps us to understand grammatical roles of words in the context.
- **Example**: Consider the sentence: "I am reading a book."
- **Tokenization**: Sentence is broken down into individual tokens or words: ["I", "am", "reading", "a", "book"]
- **Part-of-Speech Tagging**: Each token is assigned a part of speech: ["I" → Pronoun (PRP), "am" → Verb (VBP), "reading" → Verb (VBG), "a" → Article (DT), "book" → Noun (NN)]

Importance of Lexical Analysis

- **Word Identification:** It breaks text into tokens which helps the system to understand individual words for further processing.
- **Text Simplification:** It simplifies text through tokenization and stemming which improves accuracy in NLP tasks.

Morphological Analysis

It deals with **morphemes** which are the smallest units of meaning in a word. It is important for understanding structure of words and their parts by identifying free morphemes (independent words like "cat") and bound morphemes (like prefixes or suffixes e.g. "un-" or "-ing").

Key tasks in morphological analysis:

1. **Stemming**: Reducing words to their root form like "running" to "run".
2. **Lemmatization**: Converting words to their base or dictionary form considering the context like "better" becomes "good".

Importance of Morphological Analysis

1. **Understanding Word Structure:** It helps in breaking the composition of complex words.
2. **Improving Accuracy:** It enhances accuracy of tasks such as part-of-speech tagging, syntactic parsing and machine translation.

By identifying and analyzing morphemes system can identify text correctly at the most basic level which helps in more advanced NLP applications.

2. Syntactic Analysis (Parsing)

Syntactic Analysis helps in understanding how words in a sentence are arranged according to grammar rules. It ensures that the sentence follows correct grammar which makes the meaning clearer. The goal is to create a parse tree which is a diagram showing the structure of sentence. It breaks the sentence into parts like the subject, verb and object and shows how these parts are connected. This helps machines understand the relationships between words in the sentence.

Key components of syntactic analysis include:

- **POS Tagging:** Assigning parts of speech (noun, verb, adjective) to words in a sentence as discussed earlier.
- **Ambiguity Resolution:** Handling words that have multiple meanings (e.g "book" can be a noun or a verb).

Examples

Consider the following sentences:

- Correct Syntax: "John eats an apple."
- Incorrect Syntax: "Apple eats John an."

Despite using same words only the first sentence is grammatically correct and makes sense. The correct arrangement of words according to grammatical rules is what makes the sentence meaningful. By analyzing sentence structure NLP systems can better understand and generate human language. This helps in tasks like machine translation, sentiment analysis and information retrieval by making the text clearer and reducing confusion.

3. Semantic Analysis

Semantic Analysis focuses on understanding meaning behind words and sentences. It ensures that the text is not only grammatically correct but also logically coherent and contextually relevant. It aims to understand dictionary definitions of words and their usage in context and also find whether the arrangement of words in a sentence makes logical sense.

Key Tasks in Semantic Analysis

1. Named Entity Recognition (NER): It identifies and classifies entities such as names of people, locations, organizations, dates and more. These entities provide important meaning in the text and help in understanding the context. For example in the sentence "Tesla announced its new electric vehicle in California," NER would identify "Tesla" as an organization and "California" as a location.
2. Word Sense Disambiguation (WSD): Many words have multiple meanings depending on the context in which they are used. It identifies the correct meaning of a word based on its surrounding text. For example word "bank" can refer to a financial institution or the side of a river. It uses context to identify which meaning applies in a given sentence which ensures that interpretation is accurate.

4. Discourse Integration

It is the process of understanding how individual sentences or segments of text connect and relate to each other within a broader context. This phase ensures that the meaning of a text is consistent and coherent across multiple sentences or paragraphs. It is important for understanding long or complex texts where meaning focuses on previous statements.

Key aspects of discourse integration:

- **Anaphora Resolution:** Anaphora refers to the use of pronouns or other references that depend on earlier parts of the text. For example in the sentence "Taylor went to the store. She bought groceries" pronoun "She" refers back to "Taylor." It ensures that references like these are correctly understood by linking them to their antecedents.
- **Contextual References:** Many words or phrases can only be fully understood when considered in the context of following sentences. It helps in interpreting how certain words or phrases focuses on context. For example "It was a great day" is clearer when you know what event or situation is being discussed.

Example of Discourse Integration

1. "Taylor went to the store to buy some groceries. She realized she forgot her wallet."
Understanding that "Taylor" is the antecedent of "she" is important for understanding sentence's meaning.
2. "This is unfair!" helps in understand what "this" refers to we need to identify following sentences. Without context statement's meaning remains unclear.

It is important for NLP applications like machine translation, chatbots and text summarization. It ensures that meaning remains same across sentences which helps machines to understand context. This enables accurate and natural responses in applications like conversational AI and document translation.

5. Pragmatic Analysis

Pragmatic analysis helps in understanding the deeper meaning behind words and sentences by looking beyond their literal meanings. While semantic analysis looks at the direct meaning it considers the speaker's or writer's intentions, tone and context of the communication.

Key tasks in pragmatic analysis:

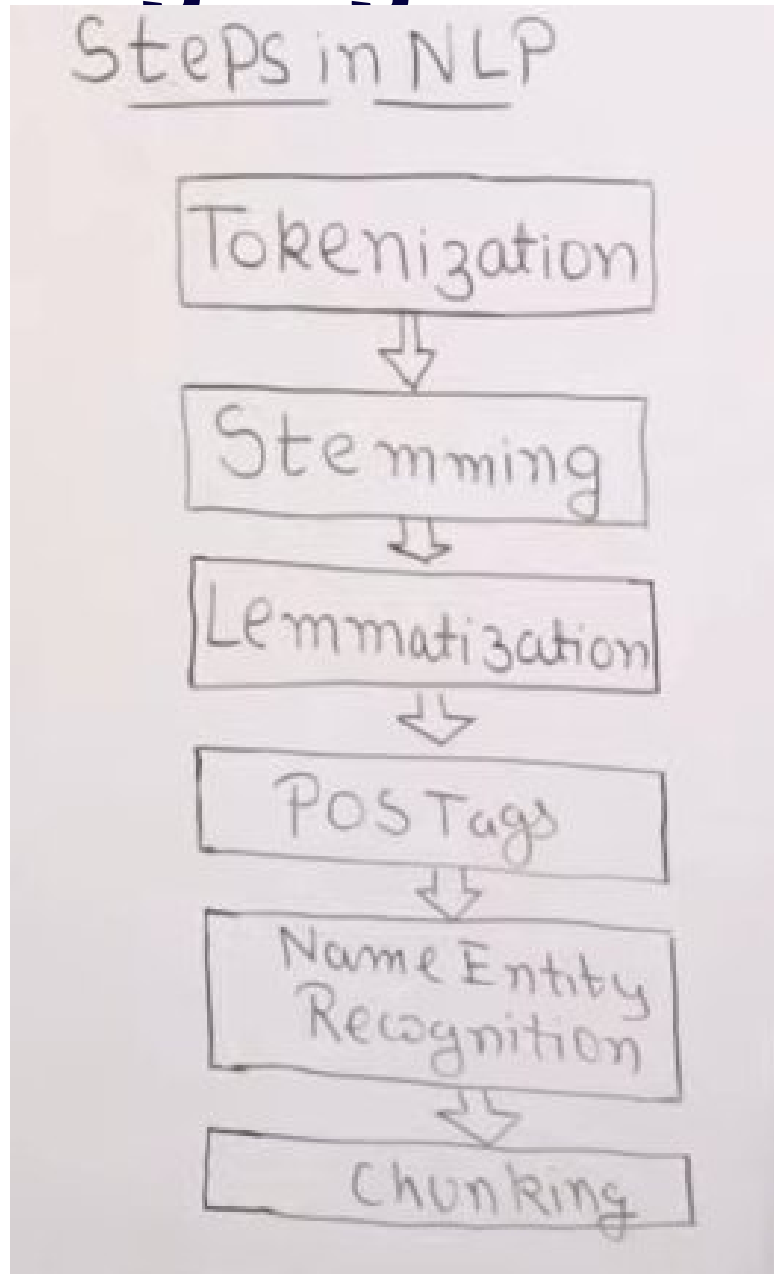
- **Understanding Intentions:** Sometimes language doesn't mean what it says literally. For example when someone asks "Can you pass the salt?" it's not about ability but a polite request. It helps to understand true intention behind such expressions.
- **Figurative Meaning:** Language often uses idioms or metaphors that can't be taken literally.

Examples of Pragmatic Analysis

- "Hello! What time is it?" here it might be a straightforward request for the current time but it could also imply concern about being late.
- For example "I'm falling for you" means "I love you" not literally falling. It helps to interpret these non-literal meanings.

- Natural language processing consists of 5 steps machines follow to analyze, categorize, and understand spoken and written language. The 5 steps of NLP rely on deep neural network-style machine learning to mimic the brain's capacity to learn and process data correctly.
- Businesses use tools and algorithms that follow the 5 NLP stages to gather insights from large data sets and make informed business decisions. Some NLP business applications include:
- Text to speech: Converting text-to-speech data, then reproducing the text as natural-sounding speech
- **Chatbots**: Helping chatbots understand and respond to customer inquiries
- **Urgency detection**: Analyzing language to prioritize tasks
- Natural language understanding: Converting speech to text and analyzing its intent
- **Autocorrect**: Detecting and removing text errors and suggesting alternatives
- **Sentiment analysis**: Revealing the perceptions people have of your

Natural Language Processing Steps

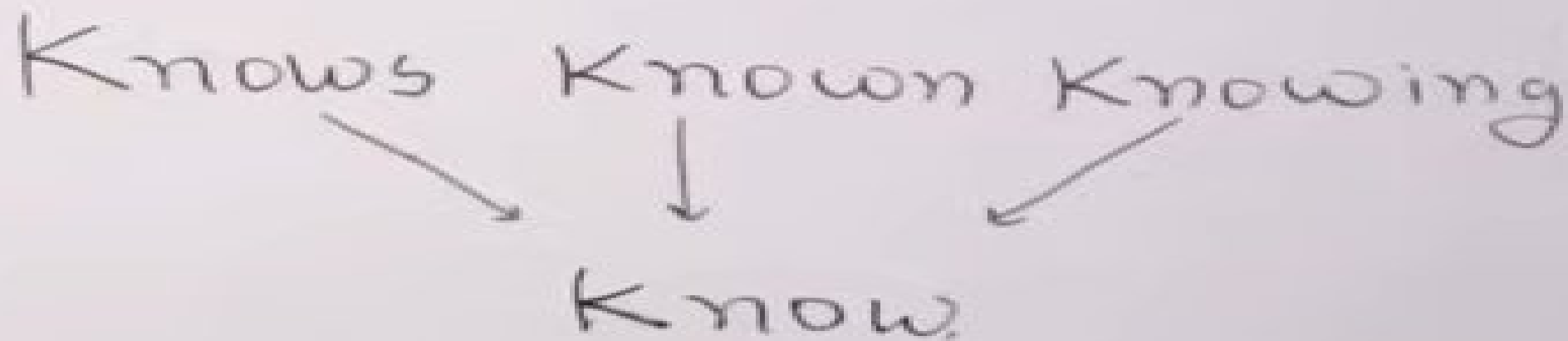


① Tokenization :

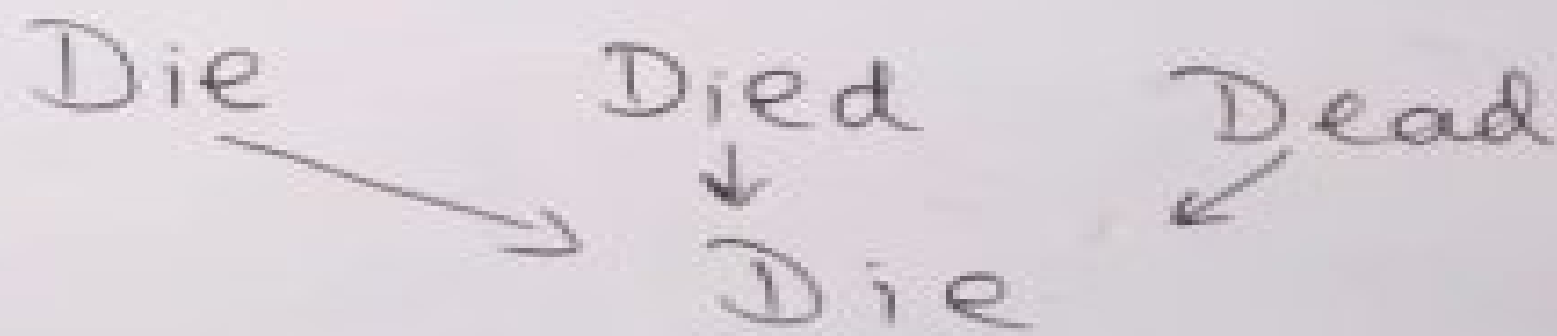
Ex: Dipesh is an abnormal human

[Dipesh] [is] [an] [abnormal] [human].

② Stemming:



③ Lemmatization



④ POS Tags

Manan Killed a bat and ate it

↓ ↓ ↓ ↓ ↓ ↓ ↓

noun verb det Noun con Verb Pron

Whatsapp me when you are free

↙ ↘

noun verb

⑤ Name Entity Recognition

Company, location, name

google it

Ques:- Explain architecture of Expert System.

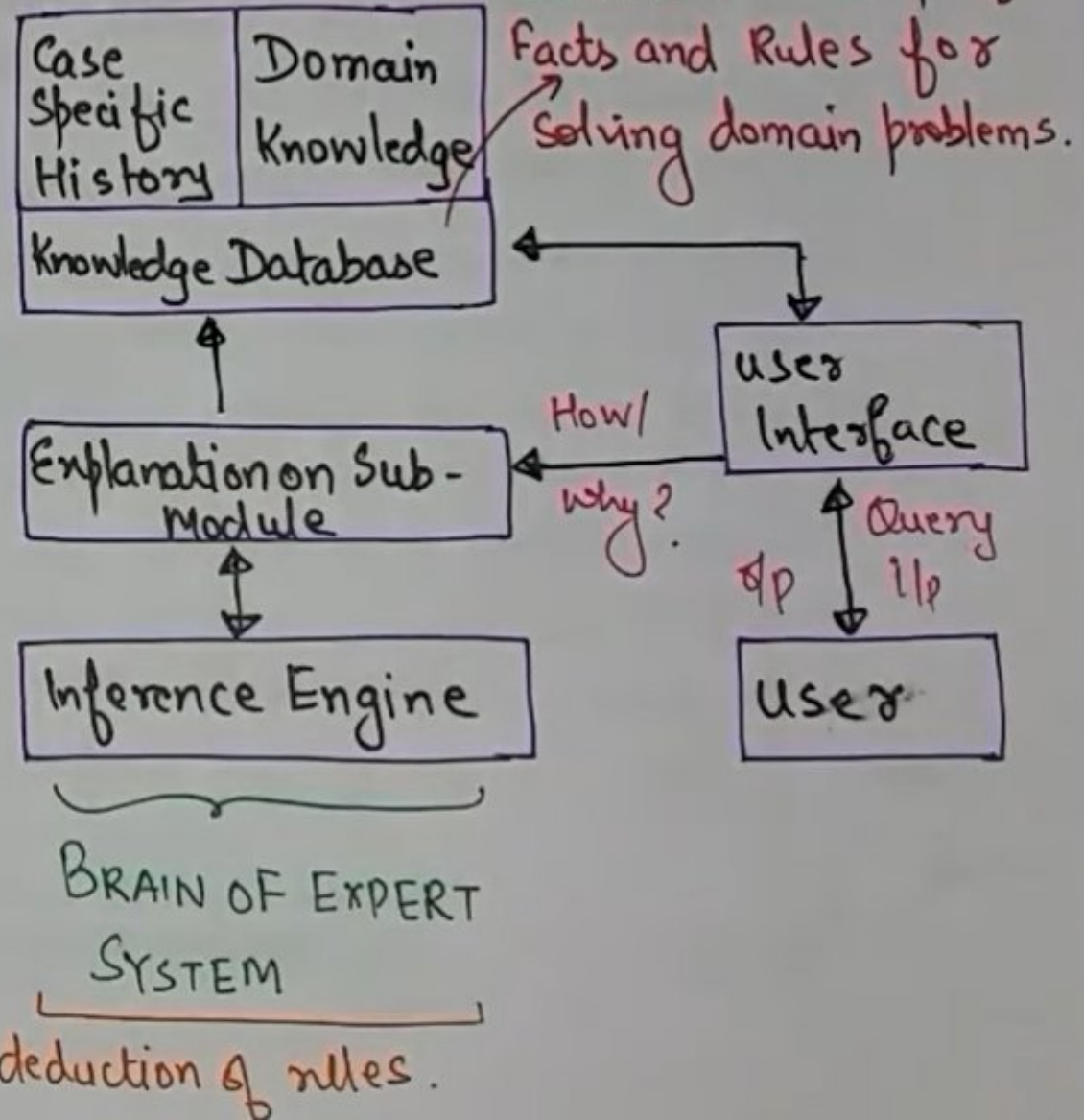
Expert System in AI are Computer Application with the level of human intelligence & expertise.

→ draws upon Knowledge of human experts Captured in a Knowledge base to solve problems that require human expertise.

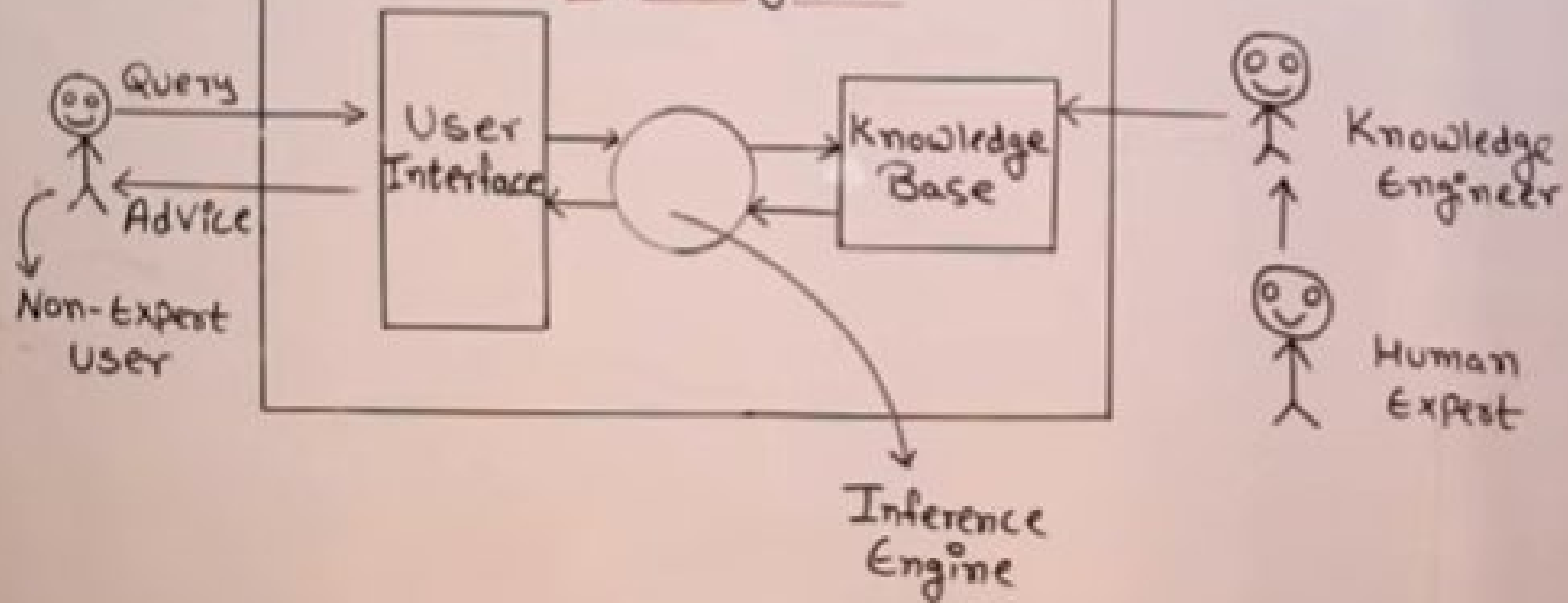
Characteristics:

- i) High Performance
- ii) Reliable
- iii) Highly Responsive
- iv) Understandable.

ARCHITECTURE:



Expert System

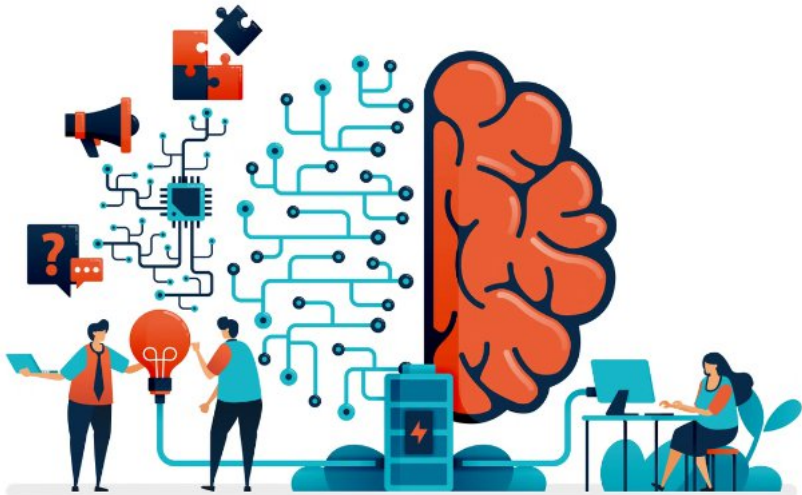


- <https://www.youtube.com/watch?v=NmF-VJ2NrT0>

1. Fully observable vs Partially Observable:

- If an agent sensor can sense or access the complete state of an environment at each point of time then it is a **fully observable** environment, else it is **partially observable**.
- A fully observable environment is easy as there is no need to maintain the internal state to keep track history of the world.
- An agent with no sensors in all environments then such an environment is called as **unobservable**.

Importance of Theorem Proving in AI



1

Formal Verification

Theorem proving is essential for verifying the correctness of software and hardware systems, ensuring their reliability and security.

2

Program Synthesis

Theorem provers can automatically generate programs from formal specifications, streamlining software development and reducing errors.

3

Automated Reasoning

Theorem provers provide a powerful tool for reasoning about complex systems, enabling AI systems to make sound and logical inferences.

4

Knowledge Representation

Theorem proving is used to represent and reason about knowledge in AI systems, facilitating intelligent decision-making and problem solving.

Logical Foundations of Theorem Proving

Propositional Logic

Propositional logic deals with propositions, which are statements that can be either true or false. It provides a basic framework for reasoning about simple logical relationships.

First-Order Logic

First-order logic extends propositional logic by introducing quantifiers, allowing reasoning about variables and objects. It provides a more expressive language for representing complex knowledge.

Higher-Order Logic

Higher-order logic allows quantification over predicates and functions, further expanding the expressiveness of the language. It is used in areas like formal verification and program synthesis.

Logical Foundations of Theorem Proving

Resolution

Resolution is a rule-based theorem proving technique that uses a set of inference rules to derive new clauses from existing ones, eventually proving or disproving the theorem.

Natural Deduction

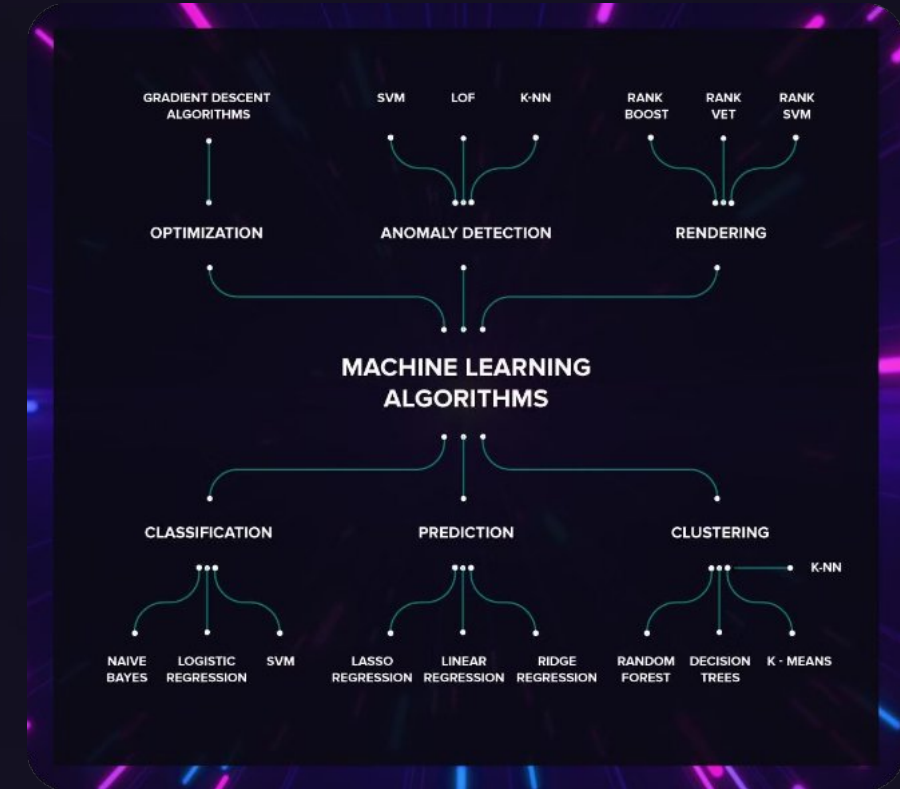
Natural deduction is a proof system that mimics human reasoning, using a set of inference rules to derive new conclusions from existing ones.

Tableau Methods

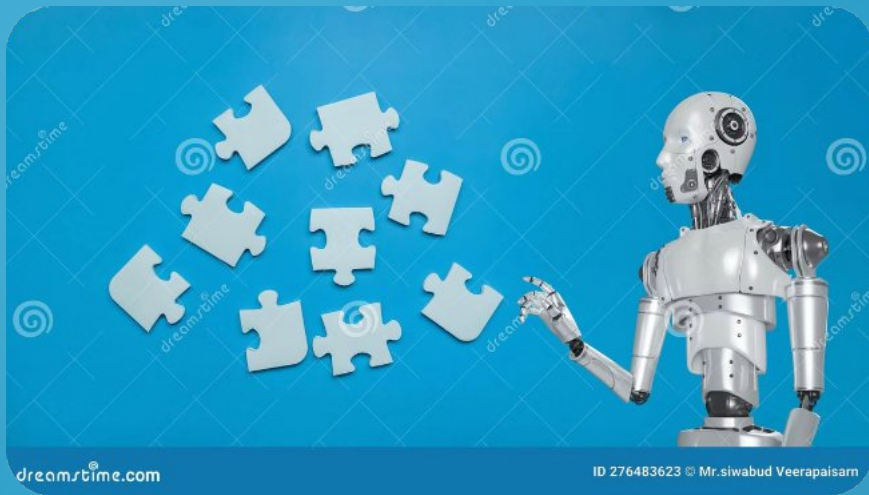
Tableau methods are based on constructing a proof tree, starting from the theorem and applying inference rules to explore possible truth values of the variables.

Model Checking

Model checking is a technique that systematically explores all possible states of a system to verify whether it satisfies a given property.



Challenges in Theorem Proving for AI



1

Computational Complexity

Theorem proving can be computationally expensive, especially for complex theorems or large systems.

2

Incomplete Information

In real-world applications, AI systems often need to reason with incomplete or uncertain information, posing challenges for theorem proving.

3

Scalability

Scaling theorem proving techniques to handle large and complex problems remains a challenge, requiring efficient algorithms and data structures.



Limitations and Potential Improvements



Limited Expressiveness

Current theorem proving techniques may struggle to represent and reason about certain complex concepts or domains.



Scalability Issues

Scaling theorem proving to handle large and complex problems remains a significant challenge.



Integration with Other AI Techniques

Integrating theorem proving with other AI techniques, such as machine learning, could enhance its capabilities and applicability.



Developing New Techniques

Exploring new theorem proving techniques and algorithms could address current limitations and improve efficiency.

Propositional Calculus:-

↳ It is a system that deals with the method used for manipulation of the symbols according to some rules.

ALPHABET SET:

↳ i) Set of Variables or Propositional Symbols P, Q, R

ii) Logical Constants $\begin{cases} \text{True (T)} \\ \text{false (F)} \end{cases}$

iii) Two Parentheses "(" and ")"

iv) Set of logical operators

IMP:-

word	symbol.	example
i) not	$\neg$	$\neg X$
ii) and	$\wedge$	$X \wedge Y$
iii) or	$\vee$	$X \vee Y$
iv) implies	$\rightarrow$	$(X \rightarrow Y)$ if X then Y
v) if and only if	$\leftrightarrow$	$(X \leftrightarrow Y)$

X : It is Hot
 Y : It is Humid
 Z : It is raining

$\left\{ \begin{array}{l} \text{All, Some} \\ \rightarrow (PL) X \end{array} \right\}$

① if it is humid then it is hot.
'Y' (X)

$\rightarrow (Y \rightarrow X)$

(X, Y)

② if it is hot and humid then it is not raining.
X Y Z

$(X \wedge Y) \rightarrow \neg Z$

v) Set of equivalence relations or laws: (P, Q, R) are Variables.

→ Commutative Laws: $P \wedge Q \cong Q \wedge P$, $P \vee Q \cong Q \vee P$

→ Associative Laws: $(P \wedge Q) \wedge R \cong P \wedge (Q \wedge R)$, $(P \vee Q) \vee R \cong P \vee (Q \vee R)$

→ Double Negation: $\sim(\sim P) \cong P$

Imp → De-Morgan's Law: $\neg(P \vee Q) \cong \neg P \wedge \neg Q$, $\neg(P \wedge Q) \cong \neg P \vee \neg Q$

→ Absorption Law: $P \wedge (P \vee Q) \cong P$, $P \vee (P \wedge Q) \cong P$

→ Law of Contradiction: $P \wedge \neg P \cong \text{false}$ $\begin{matrix} P=1 \\ \neg P=0 \end{matrix} \Big] \textcircled{0}$ $\begin{matrix} P=0 \\ \neg P=1 \end{matrix} \Big] \textcircled{0}$

→ Law of excluded middle: $P \vee \neg P \cong \text{True}$ $\begin{matrix} P=1 \\ \neg P=0 \end{matrix} \Big] \textcircled{1}$ $\begin{matrix} P=0 \\ \neg P=1 \end{matrix} \Big] \textcircled{1}$

→ Law of idempotency: $P \wedge P \cong P$ $\begin{bmatrix} P=1 \rightarrow \textcircled{1} \\ P=0 \rightarrow 0 \end{bmatrix}$

RULES OF INFERENCE:

↳ ① MODUS PONENS: If ' P ' and ' $P \rightarrow Q$ ' is given to be true, -then we can infer that ' Q ' is true.

P : It is a holiday ✓ (T)

Q : The school is closed } $\rightarrow$ we can infer that it is true.

$P \rightarrow Q$: If it is a holiday, then school is closed ✓ (T)

② MODUS TOLLENS: If ' $\sim Q$ ' and ' $\sim P \rightarrow \sim Q$ ' are given to be true, then we can infer that ' $\sim P$ ' is true.

$\sim Q$ = School is not closed.

if it is not a holiday, then school is not closed.

$(\sim P \rightarrow \sim Q)$

$\rightarrow \sim P$ } it is not a holiday (True)

Representing Simple Facts in FOPL: $\leftrightarrow$ models world in terms of objects. Defining a Sentence:-

Real-world facts can be represented as logical propositions written as well-formed formulas in propositional logic.

Symbols:- $\left. \begin{array}{l} \textcircled{1} \text{ objects} \\ \textcircled{2} \text{ Properties} \\ \textcircled{3} \text{ Relation.} \end{array} \right\} \text{Representation.}$

Symbols are formed of the following:-

$\hookrightarrow$ ① Set of all uppercase english Alphabets.

② Set of digits from 0 to 9.

③ Underscore character.

[All dogs are brown] Can't be written in propositional logic
 $\hookrightarrow$ first-order predicate logic.

Every atomic sentence is a sentence.

$\hookrightarrow$ is defined as predicate constant of arity 'n', followed by $t_1, t_2, \dots, t_n$ terms enclosed in parentheses and separated by commas.

$\hookrightarrow$ ① if 's' is sentence then $\neg s$ is sentence.

② if s_1, s_2 are sentences $\left[\begin{array}{l} s_1 \wedge s_2 \text{ (Conjunction)} \\ \checkmark \end{array} \right]$

③ " " $\left[\begin{array}{l} s_1 \vee s_2 \text{ (disjunction)} \\ \checkmark \end{array} \right]$

④ " " $\left[\begin{array}{l} s_1 \rightarrow s_2 \text{ (implication)} \\ \checkmark \end{array} \right]$

⑤ " " $\left[\begin{array}{l} s_1 \equiv s_2 \text{ (equivalence)} \\ \checkmark \end{array} \right]$

⑥ if x is Var. and s is sentence then $\forall x s$ is a sentence.

⑦ " " $\rightarrow \exists x s$ is a sentence.

Quantifiers in Predicate Calculus:-

There are two Quantifiers used in first-order predicate calculus:-

- ① Universal] for all 'x'
 $\forall x (\rightarrow)$ Such that
- ② Existential] for some 'x'
 $\exists x$ Such that

→ Constrain the meaning of sentence
containing a variable.

→ Quantifier is followed by a variable
and a sentence.

- ① All Boys like football. $\} \rightarrow \forall x: \text{Boys}(x) \rightarrow \text{Like}(x, \text{football})$
- ② Some Boys like football. $\} \rightarrow \exists x: \text{Boys}(x) \wedge \text{Like}(x, \text{football})$

Link

- <https://www.youtube.com/watch?v=HVzr8ZeTR6o>
- <https://www.youtube.com/watch?v=hHRWRa3k6Ww>
- https://www.youtube.com/watch?v=Q_15qjqX-RE
- <https://www.youtube.com/watch?v=HVzr8ZeTR6o&t=116s>